DRAFT — P1–P5 MEASURED; P4-a-block-energy HIT (excess_final=0.08727119221832025); P5 2HIT/4MISS; L-cool-dir INSUFFICIENT

Assignment 3: Optimization 2 — Experimental Report

Official handout writeup (PDF #1, ≤20 pages).
Label discipline: **MEASURED** = numbers from `media/p1/*` or `media/p2/*`; **FITTED** = curves on that grid (never "measured"); **PREDICTION** = locked pre-run bands; **HIT/MISS** only after MEASURED.
Configs / full cells → experimental appendix. Mechanism drill → intuition guide (not duplicated here).

Setup

- d8 transformer, next-token prediction; val loss ↓ better; AdamW unless overridden (P3 full-batch GD; P4(c) SGDM).
- ~614M tokens unless a helper scales down.
- Starter: `experiments/a3_optimization/` (`a3_release` @ `57c60bb`); Modal `cs312-daxton`; W&B `daxgute-stanford-university` / `assignments`.
- Batch 1 (P1): prefix → branches → measure **DONE** (30/30 cells in `media/p1/a3-mode-connectivity.json`).
- Batch 2 (P2+P3): **9/9 DONE** (`media/p2/batch2_finals.csv` / `batch2_final_metrics.json`).

Problem 1: Mode Connectivity

Config: `wsd0.0`, peak LR `0.003`, (x\in{0,50,100,250,500,1000}), (y\in{25,50,100,250,500}), `batch_order_seed`∈\{1,2\}. Details → appendix §P1.

Mechanism. Under constant post-warmup LR, late training keeps large bounce amplitude (river-valley; P2). Forking copies weights + Adam state at step (x); different batch orders then accumulate wall-orbit disagreement even when endpoint losses match. Longer shared prefix synchronizes trajectories before the fork, so midpoint weight-average stays in-basin; short (x) + long (y) is where the midpoint ridge appears.

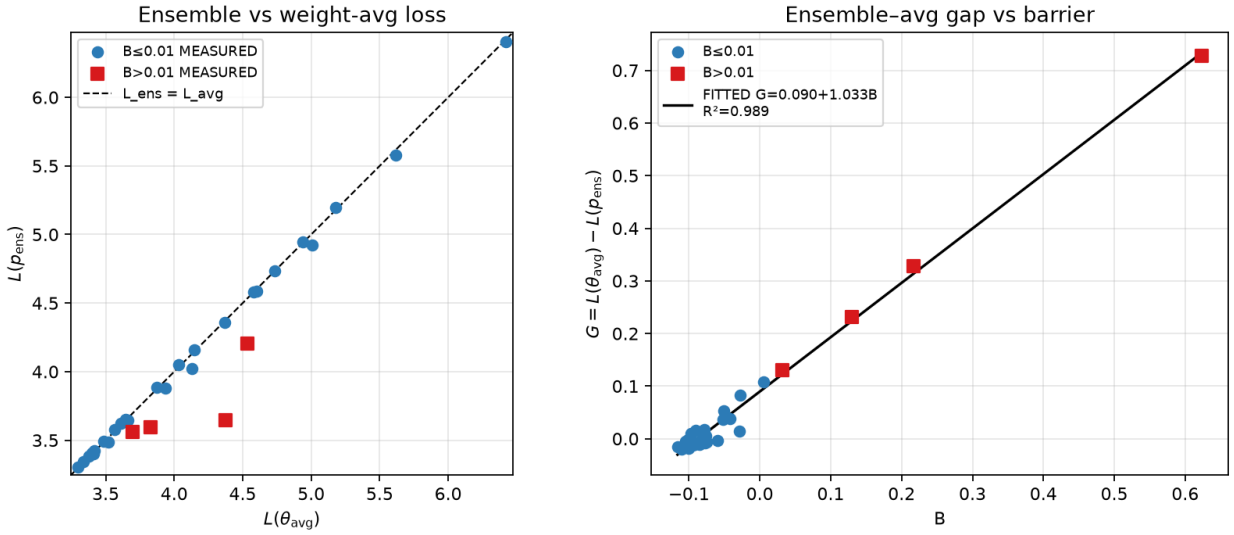
Prefix (MEASURED): `model-d8-lr0.003-tok614M-wsd0.0-stop1000` / W&B `rq1k0bz1` — end train `3.43004`, val `3.48832` (1000 steps).
Grid source: `media/p1/barrier-table.csv`, `barrier-summary.json`, `a3-mode-connectivity.json`.

(a) Endpoints + ensemble

Endpoints (L(\theta_1),L(\theta_2)) and function-space ensemble (L(p_{\mathrm{ens}})) for all 30 cells → appendix / `barrier-table.csv`.

Cell	(L_1)	(L_2)	(L_{\mathrm{ens}})	(L(\theta_{\mathrm{avg}}))
((0,500))	3.7511	3.7483	3.6454	4.3734
((1000,25))	3.4786	3.4796	3.4221	3.4194

Where (B>0.01), ensemble sits well below weight-average — parameter midpoint is bad while averaged *predictions* remain good.

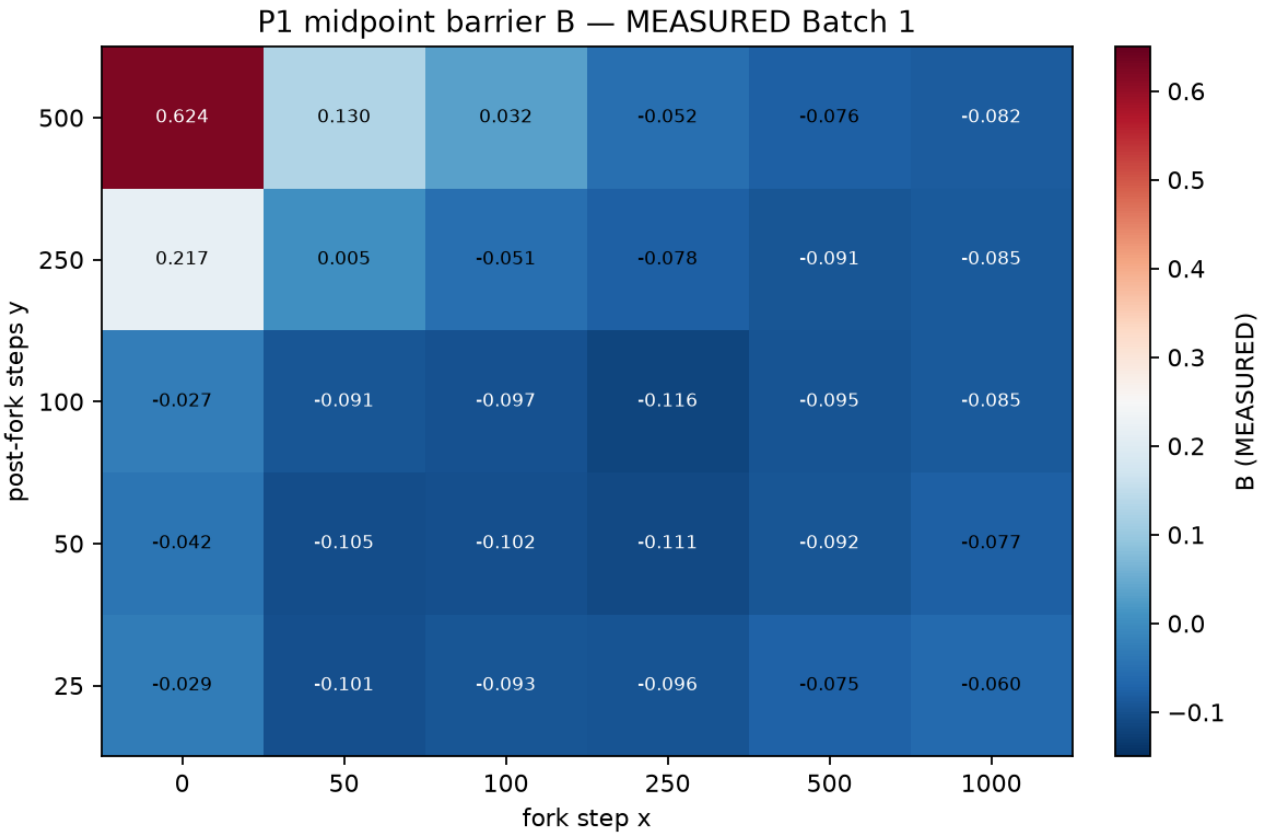


(b) Barrier (B)

(B=L(\theta_{\mathrm{avg}})-\tfrac{L(\theta_1)+L(\theta_2)}{2}) at (\alpha=0.5).

MEASURED (`barrier-summary.json`): (B_{\min}=-0.1161) @ ((250,100)); (B_{\max}=0.6237) @ ((0,500)); mean (B=-0.0334); **26/30** cells with (B\le 0.01).
Failures ((B>0.01)): ((0,250)=0.2173), ((0,500)=0.6237), ((50,500)=0.1299), ((100,500)=0.0319).
Every (x\ge 250) cell has (B\le 0.01).

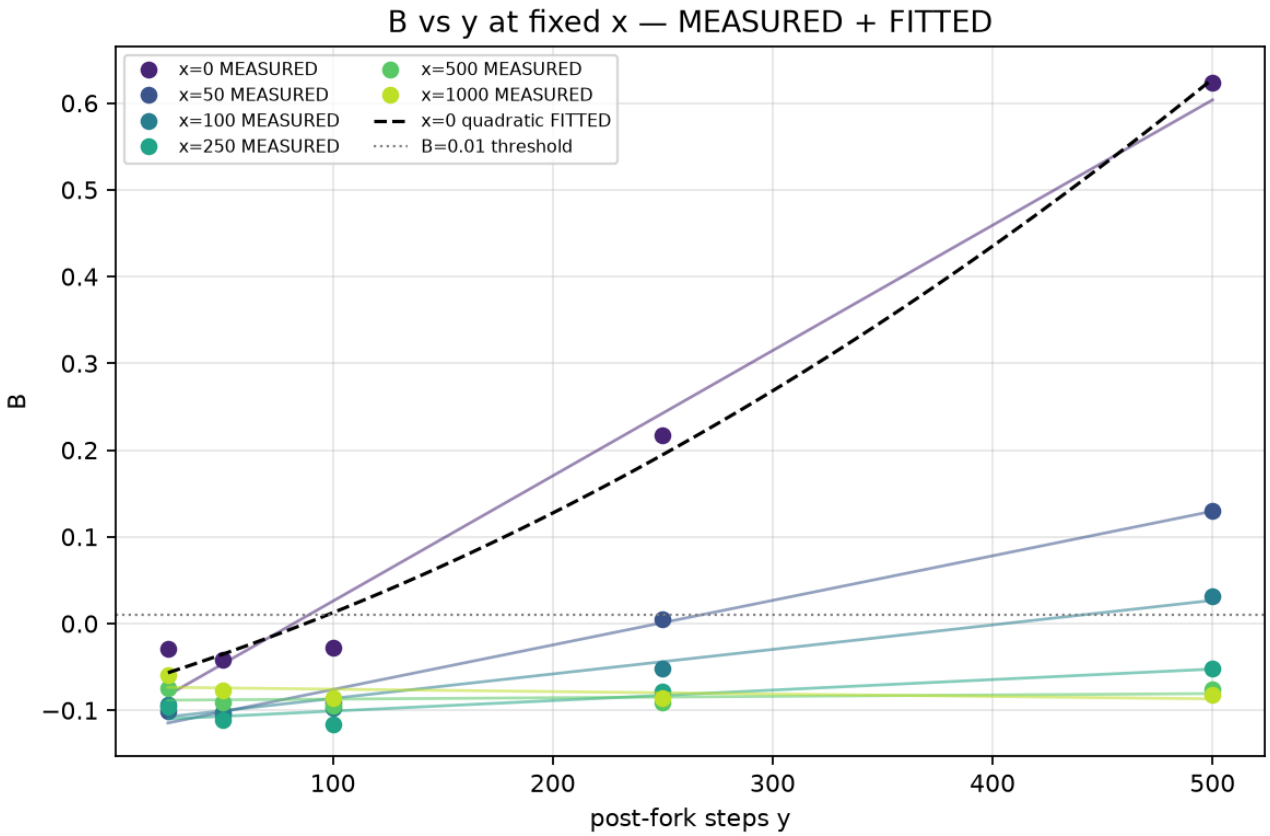
(x) \ (y)	25	50	100	250	500
0	−0.0288 ✓	−0.0416 ✓	−0.0275 ✓	0.2173	0.6237
50	−0.1009 ✓	−0.1053 ✓	−0.0910 ✓	0.0055 ✓	0.1299
100	−0.0932 ✓	−0.1019 ✓	−0.0972 ✓	−0.0512 ✓	0.0319
250	−0.0963 ✓	−0.1106 ✓	−0.1161 ✓	−0.0779 ✓	−0.0522 ✓
500	−0.0747 ✓	−0.0915 ✓	−0.0953 ✓	−0.0910 ✓	−0.0763 ✓
1000	−0.0597 ✓	−0.0769 ✓	−0.0855 ✓	−0.0855 ✓	−0.0825 ✓

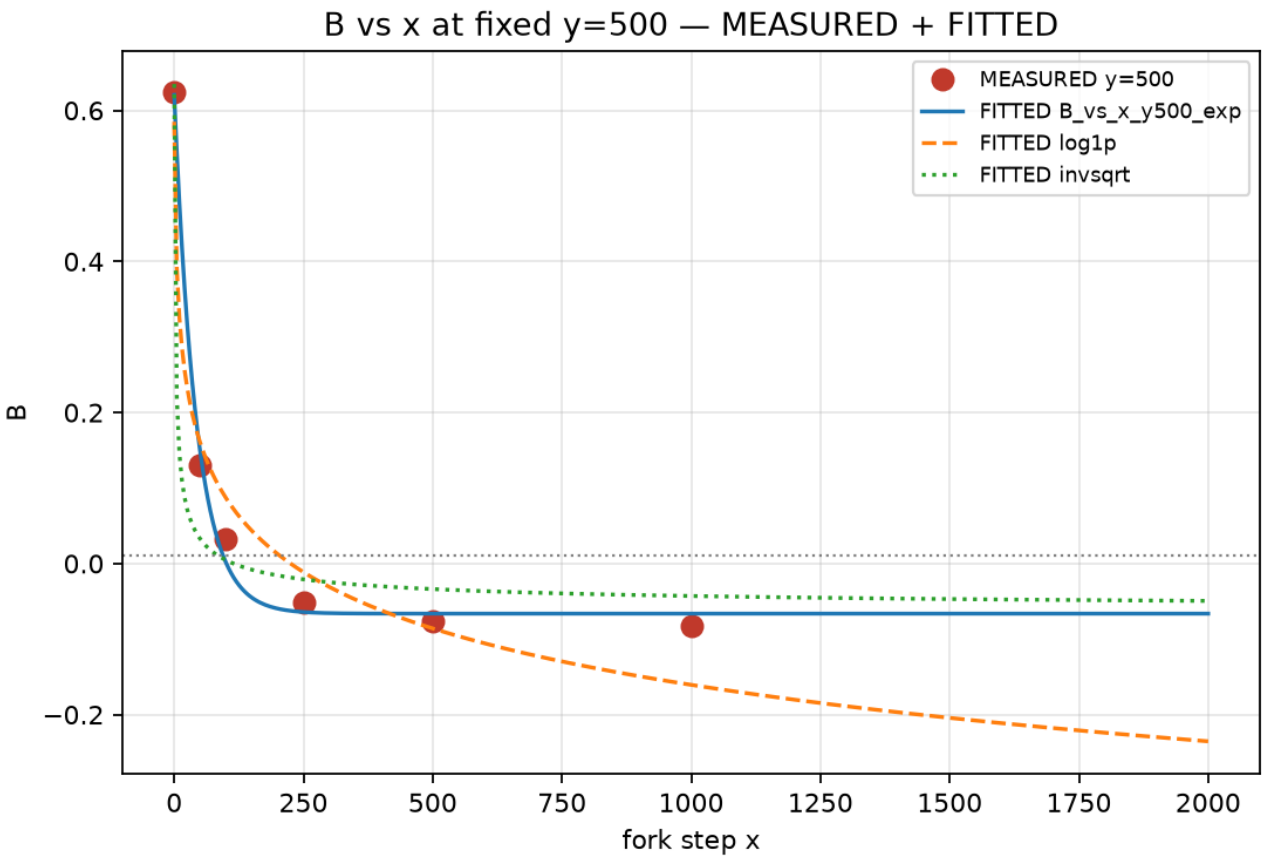


FITTED (Batch 1 grid only — not MEASURED)

ID	Relation	Form	(R^2)	Domain
Fit A growth	(B) vs (y) @ (x=0)	(B=-0.190+0.001628,y)	(\approx 1)	(x=0), (y\ge 100)
Fit B	(B) vs (x) @ (y=500)	(B=-0.066+0.687,e^{-0.0232,x})	0.995	(y=500), (x\le 1000)
Fit C	(G) vs (B)	(G=0.090+1.033,B)	0.989	all 30 cells

Identity: (G=B+(\bar L-L_{\mathrm{ens}})); MEASURED (\bar L-L_{\mathrm{ens}}) mean (\approx 0.089) ≈ constant → (G\approx B+0.09). Full coeffs → intuition-guide Appendix Q1.1 / `internal/p1-scaling-fit-status.md`.


B vs y at fixed x — MEASURED + FITTED
x=0 MEASURED
x=50 MEASURED
x=100 MEASURED
x=250 MEASURED
x=500 MEASURED
x=1000 MEASURED
x=0 quadratic FITTED
B=0.01 threshold
B
post-fork steps y
0.6
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
100
200
300
400
500



Out-of-grid bands (`P1-B-x0-y750`, `P1-B-x25-y500`, …) remain **PREDICTION** / Kind=`extrapolated` → ledger.

(c) Interpolation curves

Every cell has (\alpha\in{0,0.25,0.5,0.75,1}) in `a3-mode-connectivity.json`.

Cell	(L(\theta(\alpha)))	(B)	(L_{\mathrm{ens}})	Shape
((0,500))	[3.7511, 3.9156, 4.3734, 3.8961, 3.7483]	0.6237	3.6454	unimodal ridge @ 0.5
((50,500))	[3.7009, 3.7345, 3.8253, 3.7265, 3.6898]	0.1299	3.5943	milder ridge
((1000,500))	[3.3812, 3.3170, 3.2986, 3.3172, 3.3810]	−0.0825	3.3017	concave (no ridge)

(d) Same-basin controls

Ablations (LR / batch / schedule): **PENDING**.
Grid implication (MEASURED, `wsd0.0` only): (B>0.01) only at small (x) + large (y); Fit B excess-barrier (\sim e^{-0.023,x}) at (y=500) is the scorecard for future schedule ablations — not a Batch 2 result.

Problem 2: River-Valley Dynamics

Config / scorecard: appendix §P2 · scorecard.
MEASURED finals: `media/p2/batch2_finals.csv` (jobs 0–6). Best raw val among schedule arms: linear (L^\star=2.9282028675079346). Regret (R=L-L^\star).

Mechanism. Decayed schedules shrink late bounce amplitude and let the iterate settle into the valley floor; constant `wsd0.0` keeps wall-bouncing through the full budget, so final val sits far above decayed peers. EMA / temporary dip / horizon-free arms probe whether averaging or budget-agnostic LR can recover that gap without a tuned decay.

Job	Arm	W&B	Final val	Quantity (MEASURED)
1	linear	`r0294p93`	2.92820	(R=0) (best)
2	cos	`sktw67q8`	2.93412	(R=0.00592)
3	`wsd0.2`	`soosjcxr`	2.93774	(R=0.00954)
0	`wsd0.0`	`widt420l`	3.09751	(R=0.16930) (worst)
4	`wsd0.2`+dip	`rms7iyyt`	2.93899	(R_{\mathrm{dip}}=0.00125) vs job 3
5	invsqrt	`1jri34em`	3.02046	(\Delta L=0.09225) vs linear
6	SGDR (T=2000)	`51b3qq4n`	3.03480	(\Delta L=0.10660) vs linear

(a) Schedule comparison

PREDICTION: decayed > constant; top-2 among {`linear, cos, wsd0.2`}; `wsd0.0` (R\in[0.02,0.08)).
MEASURED ranking: linear (<) cos (<) `wsd0.2` (≪) `wsd0.0`.

Ledger	PREDICTION	Observed	
`P2-sched-rank`	(\min L_{\mathrm{decayed}} < L_{\mathrm{wsd0.0}})	linear best; `wsd0.0` worst	**HIT**
`B2-P2-linear-R`	(R\in[0,0.02))	(0)	**HIT**
`B2-P2-cos-R`	(R\in[0,0.02)); top-2	(0.00592); #2	**HIT**
`B2-P2-wsd0.2-R`	(R\in[0,0.02)); top-2	(0.00954); #3	band **HIT**; top-2 **MISS**
`B2-P2-wsd0.0-R`	(R\in[0.02,0.08)); worst	(0.16930); worst	band **MISS**; worst **HIT**

Constant-LR regret was under-predicted by ~(2\times) the upper band edge — bounce amplitude under `wsd0.0` dominates. LR/train/val plots: **PENDING** (appendix empty).

(b) Eval-only EMA

PREDICTION: gap-closed fraction (f\in[0.5,0.75)) (`P2-ema-gap-frac`).
Results: PENDING — need EMA series export; do not invent.

(c) Temporary LR dip

Dip `lr_dip=(0.4,0.45,0.5,0.1)` on `wsd0.2`; control = job 3.
PREDICTION: final (R\in[-0.02,0.02)); mid-dip (R_{\mathrm{mid}}<-0.02).
MEASURED: (R_{\mathrm{dip}}=0.00125\in[-0.02,0.02)) → `B2-P2-dip-R` **HIT** (temporary dip does not change the final). Mid-dip: **PENDING**.

(d) Horizon-free schedules

PREDICTION: (\Delta L=L_{\mathrm{hf}}-L_{\mathrm{linear}}\in[0.01,0.08)); (\lvert L_{\mathrm{sgdr}}-L_{\mathrm{invsqrt}}\rvert\in[0,0.03)).
MEASURED: invsqrt (\Delta L=0.09225) **MISS**; SGDR (\Delta L=0.10660) **MISS**; closeness (0.01435) **HIT**. Horizon-free arms track each other but both sit ~0.1 above the tuned linear floor.

Problem 3: The Edge of Stability

Config → appendix §P3. Batch 2 jobs 7–8 **MEASURED** via `media/p2/p3_*_sharpness_summary.json`.

Mechanism. Full-batch GD with fixed (\eta) drives (\lambda_{\max}) toward (\sim 2/\eta) (progressive sharpening → edge). Minibatch AdamW instead tracks a pre-update product (\eta\lambda^{\mathrm{pre}}) that can sit far above 2 while still training — a different "edge" diagnostic than SGD's (2/\eta).

(a) Full-batch (Cohen-style)

Clean case (N=32), (\eta=0.01) (job 7 / W&B `0dewkykj`). η-halve/double @ 750: **PENDING** (not in Batch 2 nine).

Quantity	PREDICTION	MEASURED	
median (\lambda_{\max}) steps 1200–1500	`[150,250)`	**245.16** (`p3_sgd_sharpness_summary.json`)	**HIT** (near (2/\eta=200))

Late train loss still decreases (final train `1.26862` @ 1499; val `8.05371` — small-(N) overfitting, not the edge metric). Crossing ~step 375; plateau (\eta\lambda\approx 2.55) (`summarize_raw`).

(b) Minibatch AdamW

Job 8 / `foew12y0` (val `3.23971` @ 2999). Primary: post-warmup median (\eta\lambda^{\mathrm{pre}}).

Quantity	PREDICTION	MEASURED	
post-warmup median (\eta\lambda^{\mathrm{pre}})	`[30,50)`	**30.99** (`p3_adamw_sharpness_summary.json`)	**HIT** (near 38)

(c) Open exploration (~18 runs)

PENDING — no Batch 2/3 exploratory arms scored here.

Problem 4: Inside the Hessian

Config → appendix §P4. Batch 3 stage a **DONE 13/13** (`ap-DAUDZsFRech705uMDN7q3x`). Full cells → `media/p4/stage-a/`; plot `media/p4/p4_lambda_max_early.png`.

Trajectory (MEASURED 13/13). Collapse by 15 → dip (\sim 2.195) @1526 (earlyfit **HIT**) → rebound to (\lambda@9375=5.967310256347902) (**inside** handout `[4.5,6.0]`). Norm mid→final **HIT**. Block-energy **HIT** (excess_final=0.08727119221832025; fam_above=True; layer_above=True). Rebound: `internal/p4-lambda-rebound-13of13.md`. Scores: `internal/p4-9375-hit-miss-summary.md`.

MISS mechanisms (predictive — locked PREDICT kept; not post-hoc rewrites). Pre-registered stories vs file-exact fails (`internal/p4-9375-hit-miss-summary.md`):

- **`P4-a-family-sharpest`:** PREDICT late sharpest ∈ {`attention`,`mlp`} (residual-block sharpening; embed/head early-only). MEASURED late=`head` ((5.641603135007765)); early_top2=`embedding`,`attention`; ratio (7.38436579244244\notin[1.5,5)). Residual-block late locus **failed** — head stayed sharpest at 9375.
- **`P4-a-density-neg`:** PREDICT neg mass **shrinks** 0→9375 (saddle cleanup) with final ∈`[0,0.15)`. MEASURED (0.004510114309467541\to\cdots\to0.06965100420936976); final band OK but shrink_0→9375=**false**. Cleanup direction **failed** on this AdamW path.
- **`P4-lam-9375-earlyfit`:** PREDICT (\lambda@9375\in[0.3,4.0)) from early decay / preferred asymptote (\sim 3.51). MEASURED (5.967310256347902\notin[0.3,4.0)) but **inside** handout `[4.5,6.0]`. Earlyfit envelope undershot the rebound into the handout edge band.

Handout finals (given): (\lambda_{\max}\in[4.5,6.0]), val (\in[2.925,2.931]). PREDICT bands unchanged.

(a) Measure provided AdamW ckpts

Results: Stage-a **MEASURED 13/13**. Copied from dry-run DRAFT_* (file-exact).

Ledger ID	PREDICTION (locked)	Observed
`P4-a-family-sharpest`	Late: attention or mlp; sharpest/median final `[1.5,5)`	**MISS** — late=`head` ((5.641603135007765)); early_top2=`embedding,attention`; ratio=(7.38436579244244)
`P4-a-norm-sharpness`	Nondecreasing; final (\lambda_{\max}) in `[4.5,6.0]`	**HIT** — (\lambda) (2.1954286466134665\to5.967310256347902); (\mathrm{tr}(H^2)/d) (8.856225326736765\mathrm{e}{-7}\to5.364973956804868\mathrm{e}{-6})
`P4-a-density-neg`	Neg mass shrinks; final neg frac `[0,0.15)`	**MISS** — neg_frac (0.00451\to0.01491\to0.93446\to0.06965100420936976); shrink_0→9375=false (final>init); final band OK
`P4-a-block-energy`	Family within-frac > random; excess `[0.05,0.30)`	**HIT** — family_within-random excess_final=0.08727119221832025; fam_above=True; layer_above=True
`P4-lam-1526-earlyfit`	(\lambda@1526\in[1.4,4.0))	**HIT** — (2.1954286466134665)
`P4-lam-9375-earlyfit`	(\lambda@9375\in[0.3,4.0))	**MISS** — (5.967310256347902) (inside handout `[4.5,6.0]`)

(b) Move (\lambda_{\max}) without moving the loss

Results: MEASURED (hand-rescale after-arm scored; see Observed).

Baseline (MEASURED `media/p4/b-rescale-reference.json`): (\lambda_{\max} =5.967310256347902), `probe_loss`(=2.9474442899227142), `scale={}` (unscaled AdamW@9375; matches stage-a final). Soft only — not an invar score.

Hand-rescale: after-arm scored from `internal/p4-b-score-dryrun.md` — `P4-invar-dL` **MISS** (|ΔL|=2.73853462934494; λ ratio=0.4658784038724411; λ 5.967310256347902→2.780040977639008; probe_loss 2.9474442899227142→5.685978919267654).

Ledger ID	PREDICTION (locked)	Observed
`P4-invar-dL`	(\lvert\Delta L_{\mathrm{val}}\rvert\in[0,0.002)); (\lambda) ratio (\ge 2) or (\le 0.5)	**MISS** — `
`P4-b-train-2x`	(\lambda) factor (\ge 2\times); val in `[2.925,2.931]`	—
`P4-b-invar-vs-train`	Hand (\lvert\Delta L\rvert) < successful train-arm (\lvert\Delta L\rvert)	—

(c) SGDM top subspace

Results: MEASURED.

Ledger ID	PREDICTION (locked)	Observed
P4-top10-frac	Mean (|P_{10}g|/|g| \in[0.6,0.85))	**MISS** — mean top_frac=0.09296940652964016
P4-c-loss-rank	full ≈ top < removed; gap [0.02,0.20)	**MISS** — L_full=3.582527816295624; L_top=5.502330541610718; L_removed=3.5821831226348877; gap=-0.00034469366073608 4
P4-c-k-ablate	Optional (k\in{3,30}) bands	—
P4-c-stale-evecs	Optional stale refresh extra loss [0,0.05)	—

Problem 5 (optional): Own prediction problem

Status: MEASURED — P5-B-cool-x0 **MISS**; P5-dB-cool **MISS**; P5-B-const-ctrl **HIT**; P5-rank-arms **MISS**; P5-G-cool-fitC **HIT**; P5-emergent-basin **MISS**; P5-L-cool-dir **INSUFFICIENT** (from media/p5/ / internal/p5-score-dryrun.md). Practice quiz **Q25**.

Phenomenon: after early fork under wsd0.0 already yields large (B), can a short matched cooldown collapse (B\le 0.01) (same valley walls) vs constant-LR control (true modes)?

Baseline (MEASURED P1 only): ((0,500)) (B=0.6237); ((50,500)) (B=0.1299); ((1000,500)) (B=-0.0825).

ID	Quantity	PREDICTION
P5-B-cool-x0	(B_C) after cooldown	((-\infty,0.01])
P5-dB-cool	(B_R-B_C) ((B_R=0.6237))	([0.5,0.8))
P5-B-const-ctrl	(B_K) after +100 constant LR	([0.5,\infty))
P5-rank-arms	Ranking	(B_C < B_R) and (B_C < B_K)
P5-emergent-basin	Forced choice	Same-valley if (B_C\le 0.01) and (B_K>0.01)

Code: experiments/a3_optimization/p5_cooldown_barrier.py. Observed: **MEASURED** — P5-B-cool-x0 **MISS**; P5-dB-cool **MISS**; P5-B-const-ctrl **HIT**; P5-rank-arms **MISS**; P5-G-cool-fitC **HIT**; P5-emergent-basin **MISS**; P5-L-cool-dir **INSUFFICIENT**. Media: media/p5/.

MISS mechanisms (predictive — locked PREDICT kept; not post-hoc rewrites). File-exact from media/p5/ / internal/p5-score-dryrun.md:

- **Cooldown did not collapse the barrier:** P5-B-cool-x0 PREDICT (B_C\le 0.01) → MEASURED B_C=0.7697049379348755 (**MISS**); P5-dB-cool PREDICT (\Delta B\in[0.5,0.8)) → dB=-0.14600493793487546 (**MISS**, negative — (B_C>B_R)); P5-rank-arms PREDICT (B_C < B_R) and (B_C < B_K) → **MISS**. Short matched cooldown failed to create connectivity on the worst P1 cell.

- **Constant-LR control stayed high-(B):** `P5-B-const-ctrl` PREDICT (B_K\ge 0.5) → B_K=0.7320129871368408 (**HIT**). Continuing constant LR did not invent a valley — control arm matches keep-bouncing / high-(B).
- **Fit C held:** `P5-G-cool-fitC` → G_C=0.8391239460323332 with |G−(B+0.09)| =0.020580991902542256 (**HIT**). Ensemble−avg gap still tracks (G\approx B+0.09) even while the barrier stays large.
- **Emergent basin:** PREDICT same-valley walls → JSON emergent='true-modes' / not-same-valley (**MISS** `P5-emergent-basin`). With (B_C) and (B_K) both ≫0.01, the forced choice is true disconnected modes, not opposite walls of one valley.
- **`P5-L-cool-dir`: INSUFFICIENT** (dL null / (L_R,L_C) missing) — no direction takeaway until endpoints land.

Interpretation summary

P1 (MEASURED + FITTED). Weight averaging works on 26/30 cells. Failures = small-(x), large-(y) under constant LR; worst (B=0.6237) at ((0,500)). There, ensemble still wins ((G) up to 0.7281). Fits A–C summarize growth in (y), exponential collapse in shared (x), and (G\approx B+0.09).

P2 (MEASURED 0–6). Decayed schedules beat constant (**HIT** rank); `wsd0.0` regret (0.169) under-predicted (**MISS** band). Dip final ≈0 (**HIT**). Horizon-free (\Delta L\sim 0.09)–(0.11) vs linear (**MISS** bands) but track each other (**HIT** closeness). EMA / mid-dip still **PENDING**.

P3 (MEASURED jobs 7–8). SGD late (\lambda_{\max}=245) → **HIT** `[150,250)`. AdamW post-warmup (\eta\lambda^{\mathrm{pre}}=31.0) → **HIT** `[30,50)`. η-change / ~18-run exploration **PENDING**.

P4 (MEASURED 13/13 + b baseline). Rebound to (\lambda@9375=5.967310256347902) (**inside** handout `[4.5,6.0]`). Predictive MISSes kept locked bands: late sharpest=`head` not attn|mlp; dens-neg no shrink 0→9375; earlyfit@9375 rebound past `[0.3,4.0)` into handout. Norm **HIT**; block-energy **HIT** (excess_final=0.08727119221832025). P4(b): reference baseline MEASURED (`scale={}`); hand-rescale scored — `P4-invar-dL` **MISS** (|ΔL|=2.73853462934494; ratio=0.4658784038724411). b/c: `P4-top10-frac` **MISS** (mean=0.09296940652964016); `P4-c-loss-rank` **MISS** (gap=-0.00034693660736084). **P5:** MEASURED — cooldown did not collapse barrier (`P5-B-cool-x0` / `P5-dB-cool` / `P5-rank-arms` / `P5-emergent-basin` **MISS**; emergent='true-modes'); constant-LR control stayed high-(B) (`P5-B-const-ctrl` **HIT** B_K=0.7320129871368408); Fit C held (`P5-G-cool-fitC` **HIT**); `P5-L-cool-dir` **INSUFFICIENT**.

Calibration: scorecard, ledger. Full tables: appendix.